From Fourier Series to Spherical Harmonics

View-Dependent Appearance in 3D Gaussian Splatting (2026/06/03)

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Section I: Background and Fundamental Theory

From ordinary signals to basis-function expansions

Background

- Fourier analysis originated from the study of heat transfer in the early 19th century.
- Joseph Fourier proposed that complex periodic signals could be represented by sums of simple sine and cosine waves.
- This idea later became a foundation of signal processing, communication systems, image analysis, and modern visual computing.
- The core insight is simple: complicated signals can be decomposed into simpler basis functions.

Complex Signal $\rightarrow$ Sine and Cosine Bases $\rightarrow$ Frequency Coefficients

Fourier Series

A periodic signal can be represented as a weighted sum of sine and cosine functions:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

where:

- a_0 represents the average level of the signal.
- a_n and b_n are Fourier coefficients.
- $\cos(nx)$ and $\sin(nx)$ are basis functions.
- n controls the frequency of each component.

Approximating a Signal by Increasing n

- Using only low-frequency terms gives a rough and smooth approximation.
- Adding more terms allows the representation to capture sharper changes and finer details.

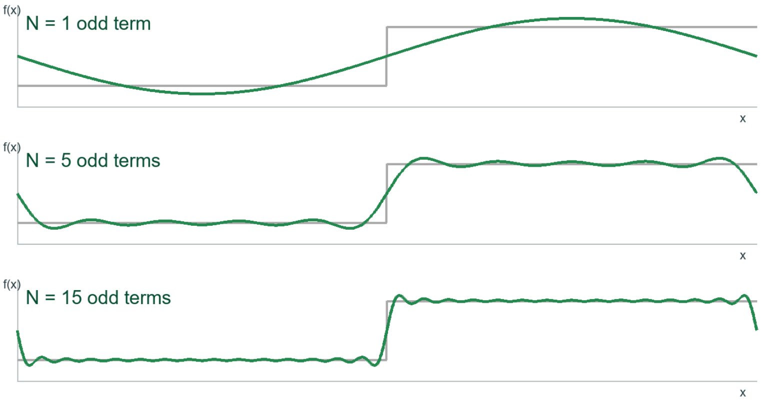


Figure: Square-wave approximation with increasing Fourier terms

Fourier Ideas in 2D Images

- For object boundaries, a closed contour can also be represented in polar form:

$$r = r(\theta)$$

- Then Fourier series can approximate this boundary by adding more frequency components.

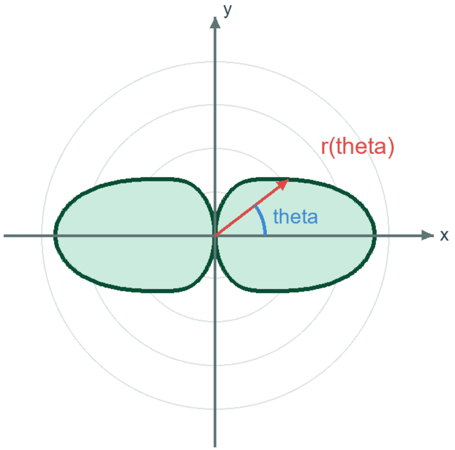


Figure: A dumbbell-shape in polar coordinates

Two-Variable Fourier Series

A two-variable signal can be written as:

$$z = f(x, y)$$

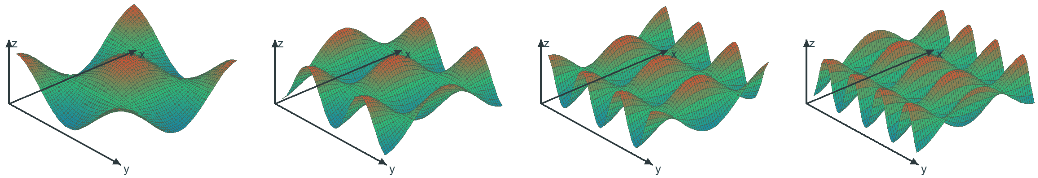
where x and y are spatial coordinates, and z is the signal value or surface height.
For a periodic 2D signal, the Fourier series can be written as:

$$f(x, y) = \sum_{m=-M}^{M} \sum_{n=-N}^{N} c_{mn} e^{j(mx+ny)}$$

Equivalently, it can be expressed using sine and cosine terms:

$$f(x, y) \approx \sum_{m=0}^{M} \sum_{n=0}^{N} \big[A_{mn} \cos(mx) \cos(ny) + B_{mn} \cos(mx) \sin(ny) \\ + C_{mn} \sin(mx) \cos(ny) + D_{mn} \sin(mx) \sin(ny)\big]$$

Basis Functions of a Two-Variable Fourier Series



(a) $\cos(x)\cos(y)$ (b) $\cos(x)\cos(2y)$ (c) $\cos(x)\cos(3y)$ (d) $\cos(x)\cos(4y)$

Figure: Basis surfaces with increasing frequency in the y direction

Two-Variable Fourier Series for Surface Signals

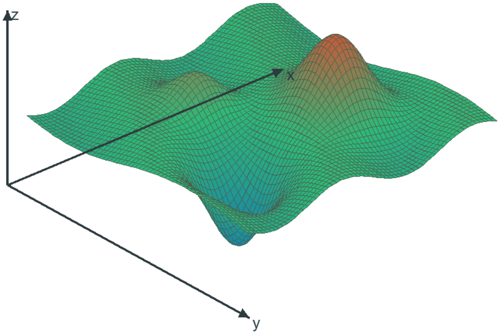


Figure: Two-variable surface signal

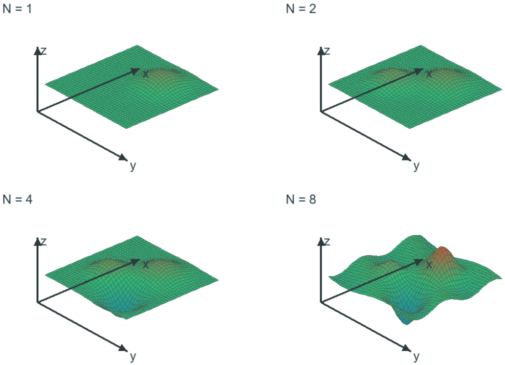


Figure: Increasing 2D Fourier terms

Section II: Spherical Harmonics

Extending Fourier representation from a plane to a sphere

Fourier-Like Expansion in Spherical Coordinates

In spherical coordinates, a 3D surface can be described by a radius function:

$$r = r(\theta, \phi)$$

where:

- r is the distance from the origin.
- θ is the polar angle.
- ϕ is the azimuth angle.

The corresponding Cartesian coordinates are:

$$x = r(\theta, \phi) \sin\theta \cos\phi$$

$$y = r(\theta, \phi) \sin\theta \sin\phi$$

$$z = r(\theta, \phi) \cos\theta$$

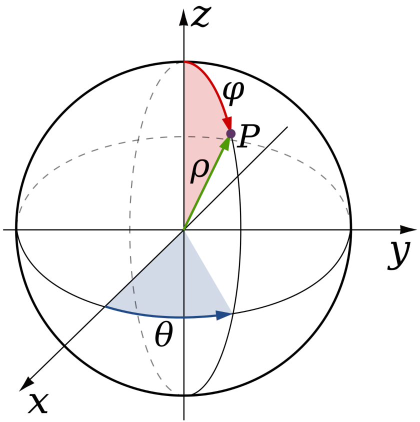


Figure: Spherical coordinates with radius, polar angle, and azimuth angle

Fourier-Like Expansion in Spherical Coordinates

Therefore, the basis expansion changes its domain from a plane to a sphere.
For a 2D signal on a plane:

$$f(x, y) \approx \sum_{m=-M}^{M} \sum_{n=-N}^{N} c_{mn} \Phi_{mn}(x, y)$$

For a directional signal on a sphere:

$$r(\theta, \phi) = \sum_{l=0}^{L} \sum_{m=-l}^{l} c_l^m Y_l^m(\theta, \phi)$$

where:

- l is the degree, controlling the frequency level.
- m is the order, controlling angular variation.
- $Y_l^m(\theta, \phi)$ is a spherical harmonic basis function.
- c_l^m is the spherical harmonic coefficient.

Spherical Harmonics and SH Coefficients

Spherical harmonics are basis functions defined on the sphere:

$$Y_l^m(\theta, \phi)$$

$$Y_l^m(\theta,\varphi) = \begin{cases} \sqrt{2}\, K_l^m \cos(m\varphi)\, P_l^m(\cos\theta), & m > 0 \\ \sqrt{2}\, K_l^m \sin(-m\varphi)\, P_l^{-m}(\cos\theta), & m < 0 \\ K_l^0 P_l^0(\cos\theta), & m = 0 \end{cases}$$

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left[(x^2 - 1)^n\right]$$

$$P_l^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} (P_l(x))$$

$$K_l^m = \sqrt{\frac{(2l+1)}{4\pi} \frac{(l - |m|)!}{(l + |m|)!}}$$

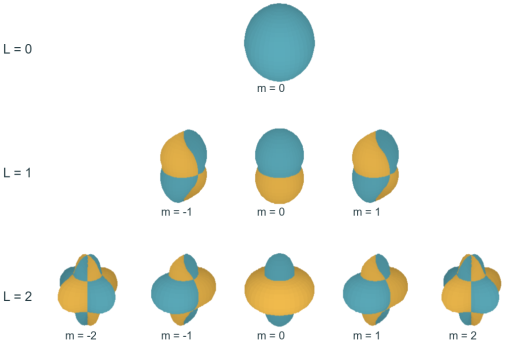


Figure: SH basis functions arranged by degree L

Spherical Harmonics and SH Coefficients

The coefficient measures how much each basis function contributes:

$$c_l^m = \int_0^{2\pi} \int_0^{\pi} f(\theta, \phi)\overline{Y_l^m(\theta, \phi)} \sin\theta \, d\theta \, d\phi$$

Figure: A signal reconstructed as weighted SH basis functions

Increasing SH Degree Improves Approximation

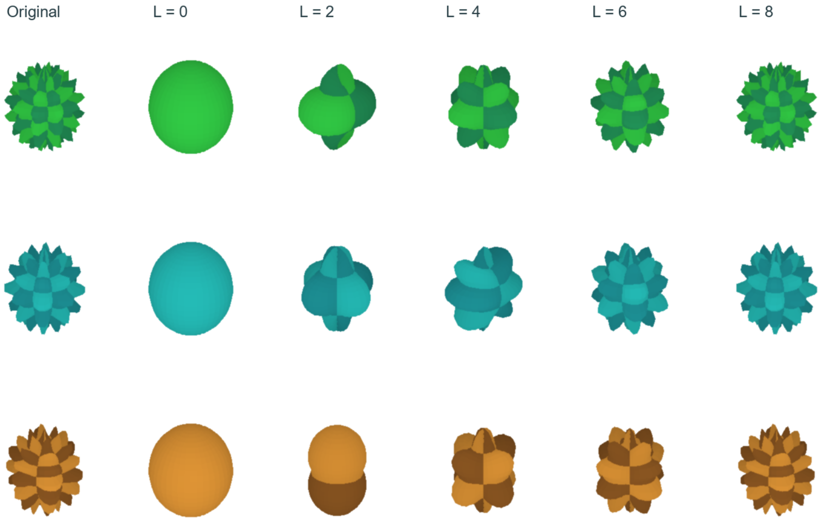


Figure: Higher SH degree captures finer angular detail

Section III: Directional Color as a Visual Signal

From color values to signal representation

Color Can Be Represented as a Signal

- Color is not only a visual property. It can also be treated as a measurable signal.
- For a fixed image point, the color value can be written as:

$$C = (R, G, B)$$

- If color changes with position, it becomes a spatial signal:

$$C(x, y) = \big(R(x, y), G(x, y), B(x, y)\big)$$

- If color changes with viewing direction, it becomes a directional signal:

$$C(\theta, \phi)$$

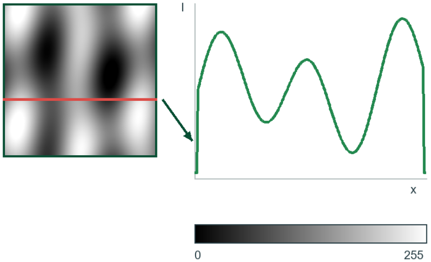


Figure: Gray values represented as a signal

From Directional Distance to Color Signal

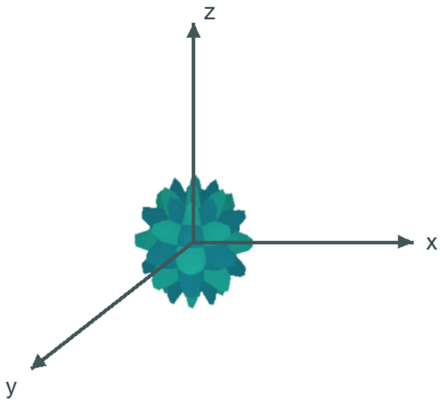


Figure: Directional distance signal

$\Longrightarrow$

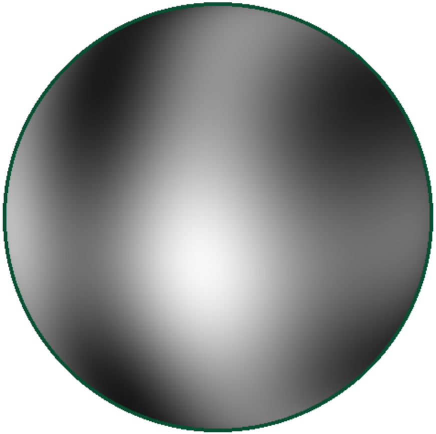


Figure: Color-mapped signal on the sphere

Multi-Channel Signals and Directional Anisotropy

- Color can be represented by three directional channels:

$$C(\theta, \phi) = \big(R(\theta, \phi), G(\theta, \phi), B(\theta, \phi)\big)$$

- Brightness, transparency, and other visual attributes can also be modeled as signals:

$$I(\theta, \phi), \quad \alpha(\theta, \phi)$$

- If a property changes with direction, it is anisotropic.

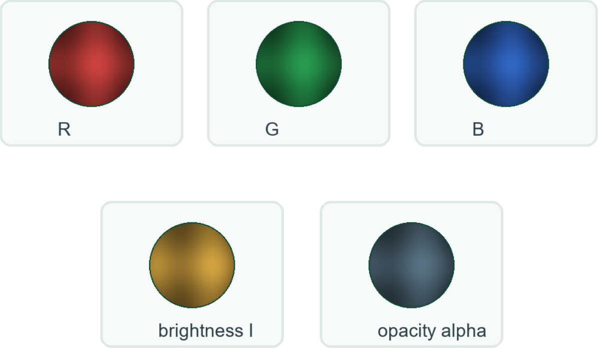


Figure: RGB, brightness, and opacity as directional signals

Section IV: Application in 3D Gaussian Splatting

Using spherical harmonics to model view-dependent appearance

SH Color Representation

Spherical harmonics are used to represent directional color:

$$\mathbf{C}_i(\boldsymbol{d}) = \sum_{l=0}^{L} \sum_{m=-l}^{l} \mathbf{c}_{i,l}^{m} Y_l^m(\boldsymbol{d})$$

where:

- $\boldsymbol{d}$ is the viewing direction from the Gaussian to the camera.
- $Y_l^m(\boldsymbol{d})$ is the SH basis value.
- $\mathbf{c}_{i,l}^{m}$ is the learned RGB coefficient.

For RGB color:

$$\mathbf{c}_{i,l}^{m} = \left(c_{R,i,l}^{m}, c_{G,i,l}^{m}, c_{B,i,l}^{m}\right)$$

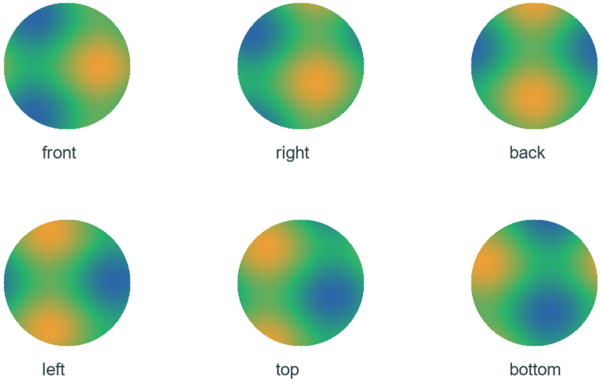


Figure: View-dependent color under SH coefficients

3D Gaussian Primitive

$$\mathcal{G}_i = \{\boldsymbol{\mu}_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}$$

where:

- $\boldsymbol{\mu}_i$ is the 3D position.
- Σ_i controls anisotropic shape.
- α_i is opacity.
- $\mathbf{c}_i$ represents color information.

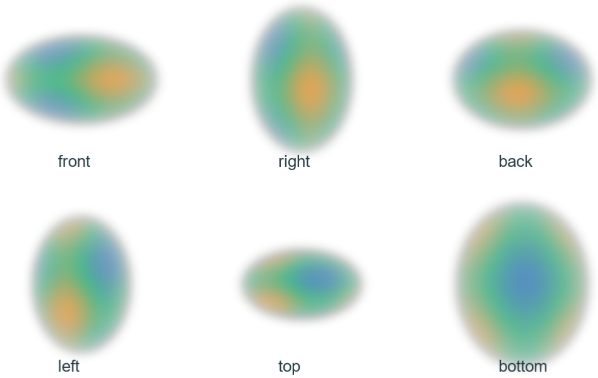


Figure: An anisotropic 3D Gaussian primitive

From One Gaussian to a 3D Scene

A single Gaussian only represents a small local primitive. By combining thousands or even millions of Gaussians, a complex object or scene can be approximated:

$$\mathcal{S} = \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_N\}$$

where each Gaussian has its own:

$$\mathcal{G}_i = \{\boldsymbol{\mu}_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}$$

During rendering, these Gaussians are projected to the image plane and blended together:

$$\mathbf{I}(u,v) \approx \sum_{i=1}^{N} T_i(u,v)\, \alpha_i'(u,v)\, \mathbf{C}_i(\boldsymbol{d})$$

where T_i represents accumulated transmittance from previously blended Gaussians.

The Pipeline of 3D Gaussian Splatting

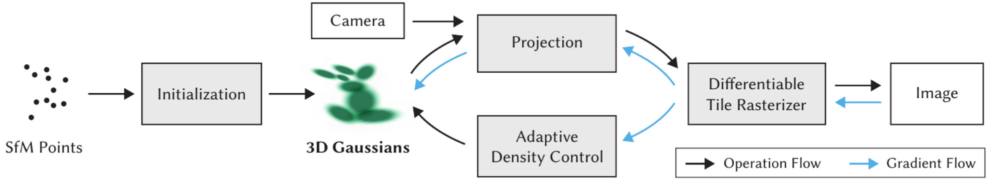


Fig. 2. Optimization starts with the sparse SfM point cloud and creates a set of 3D Gaussians. We then optimize and adaptively control the density of this set of Gaussians. During optimization we use our fast tile-based renderer, allowing competitive training times compared to SOTA fast radiance field methods. Once trained, our renderer allows real-time navigation for a wide variety of scenes.

Figure: Original optimization pipeline of 3D Gaussian Splatting

Examples of 3DGS

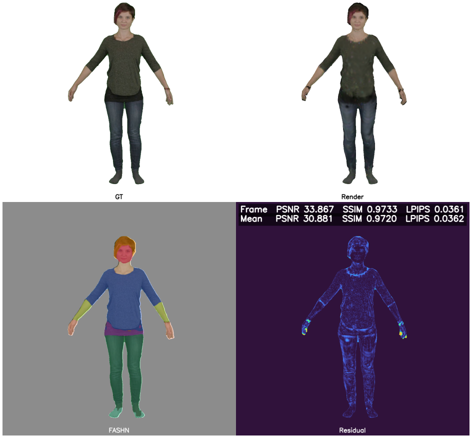


Figure: Novel view with directional color

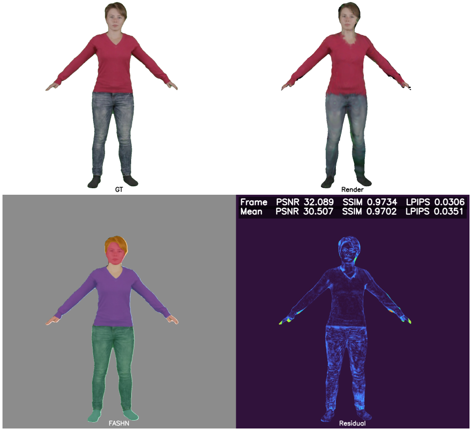


Figure: Novel view with directional color

Section V: Demo and Evaluation

A panorama-based demonstration of spherical harmonic reconstruction

Demo Design

- The user selects a 360° panorama image from a Windows desktop GUI.
- The application computes spherical harmonic coefficients for the RGB channels.
- This demo is a simplified visualization of the directional-color idea used in 3DGS.
- The demo displays the original panorama and reconstructed results under different SH orders.

Select 360° Image

$\Downarrow$

Compute SH Coefficients

$\Downarrow$

Compare Reconstructions

Conclusion

- Fourier series show how complex signals can be represented by weighted basis functions.
- Spherical harmonics extend this idea from one-dimensional or planar signals to directional signals on a sphere.
- In 3D Gaussian Splatting, SH coefficients are used to model view-dependent RGB color.
- The panorama demo illustrates how increasing the SH order improves directional signal reconstruction.

Team Members and Responsibilities

Name	Role / Contribution
Heyang Zhang	Presentation, Code Integration
Dexi Liu	Research on 3D Gaussian Splatting, Data Preparation
Shuyu Wang	Demo Implementation, Visualization
Hongwei Zeng	Analysis of Directional Color Signals, Slide Preparation
Rui Wang	Supporting Research, Literature Review, Figures Preparation